MODERN PHYSICS – MIDTERM 1.

Please fill your **Neptun code** and **Name** on **<u>all</u> papers you submit**. Preferably use these papers for all your calculations as well. If using any extra sheets of paper you submit, write your Neptun code and name on both sides of those too.
Give the answers with a precision of 3 significant digits, together with the units, unless otherwise specified. Note that if a problem is scored, e.g., X + Y + Z points, then *three* answers are expected. The later parts of every problem get progressively harder, make sure to start every problem.

Physical constants

$$c = 3.00 \times 10^{8}\,\mathrm{m/s} \qquad h = 6.626\,070\,15 \times 10^{-34}\,\mathrm{J\,s}$$
$$hc = 1.9864 \times 10^{-25}\,\mathrm{J\,m} = 1239.8\,\mathrm{eV\,nm} \qquad m_e c^2 = 511\,\mathrm{keV}$$
$$R_\infty = 1.097\,37 \times 10^{7}\,\mathrm{m}^{-1} \qquad E_0 = \frac{m_e e^4}{2\hbar^2 (4\pi\epsilon_0)^2} = 13.6\,\mathrm{eV} \quad \text{(Hydrogen ground-state energy)}$$

1. Doppler effect (4 points)

A spacecraft traveling out of the solar system at a speed of $v = 0.4\,c$ sends back information at a rate of $f_0 = 800\,\mathrm{kHz}$. At what rate f do we receive the information?

2. Leptonic decay $B^0 \to \mu^- \mu^+$ (4 + 5 + 2 + 2 points)

Rare leptonic decays provide a sensitive probe for beyond-the-standard-model effects. A B^0 meson with rest mass $m_{B^0}c^2 = 5.28\,\mathrm{GeV}$ is traveling at a speed of $v = 0.60\,c$ in the lab frame when it decays into a pair of muon μ^- and anti-muon μ^+, both with rest mass $m_\mu c^2 = 105\,\mathrm{MeV}$.

(a) What is the *total* kinetic energy K of the two muons in the *rest frame* of the decaying meson?

(b) What is the speed of the muons u' in the *rest frame* of the decaying meson?

(c) What are (i) the highest speed $u_{\max}$ and (ii) the lowest speeds $u_{\min}$ the muons may have in the *lab frame*?

Hint: Keep track of at least four significant digits to appreciate the difference between the highest and lowest speeds.

3. Compton scattering (3 + 3 + 4 points)

A photon with energy $E = 40\,\mathrm{keV}$ scatters from a free electron at rest.

(a) What is the wavelength λ of the incident photon?

(b) If the scattering angle is $\theta = 60^\circ$, calculate the *red-shifted* wavelength λ' of the scattered photon.

(c) What is the maximum recoil kinetic energy (over all scattering angles) the electron can obtain in the process?

4. Absorption and emission processes in the Hydrogen atom (4 + 4 points)

A hydrogen atom in an excited state (i.e., $n_i > 1$) absorbs a photon of wavelength $\lambda = 486\,\mathrm{nm}$, and is excited to the state n_f. After some time, it decays to the ground state $n = 1$, emitting a photon with energy E_γ.

(a) What were the initial (n_i) and final (n_f) states corresponding to the absorption process?

(b) What is the energy E_γ of the emitted photon?

Hint: Find a rational approximation with a square in the denominator. Use the infinite nucleus mass approximation.

5. Particle in a box (2 + 2 + 3 + 3 + 5 points)

A particle of mass m is confined to an infinite potential well in the $[0, L]$ interval. Given the *trial* wavefunctions $\psi_n(x) = A\cos(\pi n x/L + \phi)$ for $n \in \mathbb{N}^+$

(a) Find (i) the phase ϕ and (ii) the normalization constant A, so that $\psi_n(x)$ are normalized solutions of the time-independent Schrödinger equation.

(b) Sketch the probability density function and calculate the probability of finding the particle in the $0 < x < L/4$ interval for (i) ψ_1 and (ii) ψ_2.

(c) What is the momentum uncertainty $\Delta p = \sqrt{\langle p^2\rangle - \langle p\rangle^2}$ for state ψ_n?

Give the answers parametrically using L, m, and n if necessary.

OFFICIAL SOLUTIONS

1. Doppler effect (5 points)
A spacecraft traveling out of the solar system at a speed of $v = 0.4\,c$ sends back information at a rate of $f_0 = 800\,\mathrm{kHz}$. At what rate f do we receive the information?

Solution
The frequency f perceived by an observer on Earth is given by

$$f = f_0 \sqrt{\frac{1+\beta}{1-\beta}}.$$

where $\beta < 0$ since the source and the observer on Earth are receding. So for $\beta = -v/c$ it yields $f = 524\,\mathrm{kHz}$.

2. Leptonic decay $B^0 \to \mu^- \mu^+$ (4 + 5 + 2 + 2 points)
Rare leptonic decays provide a sensitive probe for beyond-the-standard-model effects. A B^0 meson with rest mass $m_{B^0}c^2 = 5.28\,\mathrm{GeV}$ is traveling at a speed of $v = 0.60\,c$ in the lab frame when it decays into a pair of muon μ^- and anti-muon μ^+, both with rest mass $m_\mu c^2 = 105\,\mathrm{MeV}$.

(a) What is the *total* kinetic energy K of the two muons in the *rest frame* of the decaying meson?

(b) What is the speed of the muons u' in the *rest frame* of the decaying meson?

(c) What are (i) the highest speed $u_{\max}$ and (ii) the lowest speeds $u_{\min}$ the muons may have in the *lab frame*?

Hint: Keep track of at least four significant digits to appreciate the difference between the highest and lowest speeds.

Solution
(a) In the frame of the decaying B^0 meson, the total energy is

$$E = E_{0\,B^0} = 5.28\,\mathrm{GeV}$$

and since the energy is conserved in the decay process

$$E = K + 2E_{0\mu}$$

so that the difference of rest masses is transformed into kinetic energy available for the muonic pair

$$K = E_{0\,B^0} - 2\,E_{0\mu} = 5.07\,\mathrm{GeV}$$

(b) In that reference frame, the μ^- and μ^+ particles must recoil with equal momenta in opposite directions in order to conserve momentum. Hence, the kinetic energy of each muon is

$$K_\mu = K/2 = 2.535\,\mathrm{GeV}.$$

The speed of the muon can be found by calculating γ

$$\gamma = \frac{E_{0\mu} + K_\mu}{E_{0\mu}} = 25.14,$$

so that

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = 0.9992$$

or $u' = 0.9992\,c$. Note that the symbol u' was used to emphasize that the velocity is obtained in the reference frame co-moving with the meson.

(c) The highest and lowest speeds in the lab frame are obtained when the muons are released in the forward and backward directions with respect to the moving meson, respectively. Then by using the velocity addition laws

$$u_{\max} = \frac{u' + v}{1 + \dfrac{v}{c^2}u'} = 0.9998\,c \quad u_{\min} = \frac{u' - v}{1 - \dfrac{v}{c^2}u'} = 0.9978\,c.$$

3. Compton scattering (3 + 3 + 4 points)
A photon with energy $E = 40\,\mathrm{keV}$ scatters from a free electron at rest.

(a) What is the wavelength λ of the incident photon?

(b) If the scattering angle is $\theta = 60°$, calculate the *red-shifted* wavelength λ' of the scattered photon.

(c) What is the maximum recoil kinetic energy (over all scattering angles) the electron can obtain in the process?

Solution
(a) The wavelength associated with the photon is given by

$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{E} = 0.0310\,\mathrm{nm}$$

(b) The change in the wavelength due to Compton scattering is given by

$$\lambda' - \lambda = \frac{h}{m_e c}\left(1 - \cos(\theta)\right)$$

so that for $\theta = 60°$ one finds

$$\lambda' - \lambda = \frac{h}{m_e c}\frac{1}{2} = \frac{hc}{m_e c^2}\frac{1}{2} = 0.001\,21\,\mathrm{nm}$$

Since $\lambda' > \lambda$ (always) $f' < f$, and the photon is indeed *red-shifted* with $\lambda' = 0.0322\,\mathrm{nm}$

(c) The total energy of the systems $e^- + \gamma$ before the scattering is given by

$$E = hf + m_e c^2$$

and after the scattering is given by

$$E' = hf' + m_e c^2 + K$$

Hence, by the conservation of energy $E = E'$, and the recoil kinetic energy of the electron is therefore equal to the energy lost by the photon

$$K = h(f - f')$$

Hence, the recoil kinetic energy is maximized when the photon is *back-scattered*, i.e., for $\theta = \pi$, for which

$$\lambda' - \lambda = \frac{hc}{m_e c^2}\,2$$

corresponding to $\lambda' = 0.035\,85\,\mathrm{nm}$

$$\Delta E_\gamma = h(f - f') = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right)$$

resulting in a kinetic energy

$$K = 5.414\,\mathrm{keV}.$$

4. Absorption and emission processes in the Hydrogen atom (4 + 4 points)
A hydrogen atom in an excited state (i.e., $n_i > 1$) absorbs a photon of wavelength $\lambda = 486\,\mathrm{nm}$, and is excited to the state n_f. After some time, it decays to the ground state $n = 1$, emitting a photon with energy E_γ.

(a) What were the initial (n_i) and final (n_f) states corresponding to the absorption process?

(b) What is the energy E_γ of the emitted photon?

Hint: Find a rational approximation with a square in the denominator. Use the infinite nucleus mass approximation.

Solution
(a) The absorption process is characterized by the Rydberg formula

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

and the given wavelength (in the red region of the visible range) corresponds to a (Balmer) transition between the states labeled by $n_1 = 2$ and $n_f = 4$.

(b) When decaying to the ground state, the Hydrogen atom emits a photon with energy

$$E_\gamma = hf = E_0 \left(1 - \frac{1}{n_f^2} \right) = 12.75\,\mathrm{eV}$$

where $E_0 = 13.6\,\mathrm{eV}$ is the ground-state binding energy of the Hydrogen atom. It can be verified that it is a transition in the Lyman series with $\lambda = 97.3\,\mathrm{nm}$ (UV range).

5. Particle in a box (2 + 2 + 3 + 3 + 5 points)
A particle of mass m is confined to an infinite potential well in the $[0, L]$ interval. Given the *trial* wavefunction $\psi_n(x) = A\cos(\pi n x/L + \phi)$ with $n \in \mathbb{N}^+$

(a) Find (i) the phase ϕ and (ii) the normalization constant A, so that $\psi_n(x)$ is a normalized solution of the time-independent Schrödinger equation.

(b) What is the probability of finding the particle in the $0 < x < L/4$ interval for (i) ψ_1 and (ii) ψ_2?

(c) What is the momentum uncertainty $\Delta p = \sqrt{\langle p^2\rangle - \langle p\rangle^2}$ for state ψ_n?

Give the answers parametrically using L, m, and n if necessary.

Solution
(a) The phase can be fixed requiring that the trial function fulfills the boundary conditions

$$\psi_n(x=0) = \psi_n(x=L) = 0$$

e.g., for $x = 0$ one finds

$$\psi_n(x=0) = 0 = A\cos(\phi)$$

and since $A \neq 0$, it yields $\phi = \pi/2 + m\pi$, with $m \in \mathbb{N}$, and choosing, e.g., $m = 1$ ($\phi = 3\pi/2$) the trial wavefunction reads

$$\psi_n(x) = A\cos(n\pi x/L + \phi) = \cos(n\pi x/L)\cos(3\pi/2) - \sin(n\pi x/L)\sin(3\pi/2) = A\sin(n\pi x/L),$$

and it can be verified that the other boundary condition is also fulfilled. The normalization condition is

$$1 = |A|^2 \int_0^L \mathrm{d}x |\psi_n(x)|^2 = |A|^2 \frac{L}{2n\pi} \int_0^{n\pi} \mathrm{d}y \left[1 - \cos(2n\pi x/L)\right],$$

obtained using trigonometric identities and the substitution $y = (n\pi/L)\,x$ (cfr. tutorial notes). Then the first term integrates to $n\pi$ and the second term integrates to zero (by symmetry), thus yielding for any n

$$A = \sqrt{\frac{2}{L}}.$$

(b) The probability of finding the particle in the interval $0 \leq x \leq L/4$ is obtained by solving the corresponding integral with the same strategy. For the ground state wavefunction $\psi_1(x)$ is given by

$$P_1(0 \leq x \leq L/4) = \frac{2}{L}\int_0^{L/4} \mathrm{d}x |\psi_1(x)|^2 = \frac{2}{L}\int_0^{L/4} \mathrm{d}x \sin^2(\pi x/L) = \frac{\pi - 2}{4\pi} \approx 0.0908.$$

Here we can use the trigonometric identity $\sin^2 x = (1 - \cos 2x)/2$.
For the first excited state wavefunction $\psi_2(x)$ is given by

$$P_2(0 \leq x \leq L/4) = \frac{2}{L}\int_0^{L/4} \mathrm{d}x |\psi_2(x)|^2 = \frac{2}{L}\int_0^{L/4} \mathrm{d}x \sin^2(\pi x/L) = \frac{1}{4}.$$

This result can also be obtained by looking at the probability density function and using its symmetries.
(c) The uncertainty on the momentum is given by

$$\Delta p = \sqrt{\langle p^2\rangle - \langle p\rangle^2}$$

so for a generic eigenstate wavefunction $\psi_n(x)$, recalling the operatorial definition of the momentum $p = -i\hbar\dfrac{d}{dx}$, one needs to calculate the following integrals

$$\langle p\rangle = \frac{\hbar}{i}\frac{2}{L}\int_0^L \mathrm{d}x \sin(n\pi x/L)\,\frac{d}{dx}\sin(n\pi x/L) = \frac{\hbar}{i}\frac{2n\pi}{L^2}\int_0^L \mathrm{d}x \sin(n\pi x/L)\cos(n\pi x/L) = 0$$

which is zero by symmetry, and

$$\left\langle p^2\right\rangle = -\hbar^2 \frac{2}{L}\int_0^L \mathrm{d}x \sin(n\pi x/L)\,\frac{d^2}{dx^2}\sin(n\pi x/L) = \frac{\hbar^2\pi^2}{L^2}\,n^2\frac{2}{L}\int_0^L \mathrm{d}x \sin^2(n\pi x/L) = \frac{\hbar^2\pi^2}{L^2}\,n^2$$

which after the derivative gives an integral identical to that of the normalization condition.

Finally, the momentum uncertainty for $\psi_n(x)$ is given by

$$\Delta p = \frac{\hbar\pi}{L}\,n.$$